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How manipulation of playing area dimensions in ball possession games constrains physical effort and technical actions in under-11, under-15 and under-23 soccer players

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ABSTRACT

The aim of this study was to investigate the effects of playing area manipulation (20 × 15 m, 25 × 20 m and 30 × 25 m) on external workloads (total distance covered, distance covered while walking, running and sprinting, number of sprints, maximum sprint speed), internal load perceptions (rating of perceived exertion) and technical actions of passing (number of passes with dominant and non-dominant foot, and maximum passing speed) during 4v4 ball possession small-sided and conditioned games in under-11, under-15 and under-23 soccer players. Results showed higher values in the large playing area for under-11 in the distance covered in different speed zones, sprint number and RPE (all $p < .001$) for under-15 in sprints number ($p < .01$) and maximum sprint speed ($p = .02$), and for under-23 in both RPE and sprint numbers ($p < .01$). Although no significant differences were found on technical actions, it was still possible to notice some effects through pairwise comparison. High-intensity running was promoted on larger playing areas, where under-11 s were also able to perform more technical actions of passing. Opposite, under-23s were able to perform more passing on smaller playing areas, where under-11 s perceived the exercise more intense. The impact of different playing areas was reduced for the under-15.

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Constraint led approach; task constraints; design of practice tasks; coaching development; small-sided and conditioned games

Introduction

Past research has been investigating how manipulation of constraints in team sports training influences performance (Olthof et al., 2018). Specifically in soccer, coaches manipulate training constraints on a daily basis, whether in the lab (e.g. isokinetic assessments), gym (e.g. rehabilitation) or at the pitch (e.g. small-sided and conditioned games, SSCGs) (Ibáñez et al., 2019). Therefore, it is necessary to understand the nature of

those manipulations, as their correct manipulation allows to achieve the intended intensity, and simultaneously develop technical-tactical skills under decision-making processes (Alves et al., 2017). The constraints-led approach (CLA) argues that players' and teams' behaviour emerges under the influence of interacting constraints such as the participants, the task and the environment (Davids et al., 2016). From a practical point of view, the manipulation of constraints permit to emphasize specific informational variables that attract players to a specific individual and collective patterns of play (Davids et al., 2013). By changing task constraints during SSCGs coaches may directly amplify or inhibit the range of players' action possibilities within the context of practice (Chow et al., 2016), where the playing area manipulation represents the variable which most influence players' external workload and technical performance, and consequently, decision-making (Sangnier et al., 2019).

That is, SSCGs allow coaches to promote the development of players' technical and tactical skills, while at the same time improving physiological parameters such as strength, agility and endurance in order to simulate the demands of competitive performance (Sgrò et al., 2018). However, to achieve that aim, there is a need to understand the differential effects of specific practice task constraints on the technical, tactical and physical performance of players and teams (Sarmiento et al., 2018). Past research consistently demonstrated the potential of the manipulation of constraints in SSCGs to promote specific interpersonal patterns of coordination (e.g. Silva et al., 2014; Vilar et al., 2014), develop specific space–time relationships on the use of space and ball possession (e.g. Coutinho et al., 2019; Travassos et al., 2018) and target specific effort regimes (e.g. Casamichana & Castellano, 2010; Clemente et al., 2014). For instance, it was observed that playing area dimensions influence the intensity of the game, the actions of the players and the energy systems used (Sangnier et al., 2019): large playing areas are associated with an increase in the intensity of exercise (Halouani et al., 2014; Sarmiento et al., 2018) effective playing space and surface coverage (Silva et al., 2014); while small playing areas appear to foster technical development (Clemente et al., 2019; Sgrò et al., 2018).

Despite the empirical evidence associated to the manipulation of playing area and format in SSCGs, little is known about their modulating effects on players' performance across different age groups and expertise levels (Lemes et al., 2019). This question is of great interest since player's behaviour is considered to depend on their individual capabilities and how they interact with the context to perceive individual and collective possibilities for action. Accordingly, age and experience of the players may directly influence how players perceive and explore action possibilities for themselves and their counterparts in a specific performance context (Menuchi et al., 2018). For example, a previous study has showed that younger players tend to centre their attention on the ball and solve game tasks individually by approaching the ball instead of seeking a collective solution, which has been explained by a lower level of perceptual attunement to key information variables and lower individual capability to perform (Folgado et al., 2014). It means that SSCGs need to be designed according to players' individual capabilities and more research is needed to understand the acute effect of playing area dimension manipulations in SSCGs across different age groups. This information may critically enlighten the development of dedicated intervention programmes aimed at enhancing

performance and transferring skills from practice environments to competitive settings according to players' needs (Barnabe et al., 2016; Travassos et al., 2018).

Thus, the aim of this study was then to evaluate the acute effects of playing area manipulations in ball possession SSCGs on physical effort and technical actions of players across three different age groups (U11, U15 and U23). Specifically, it sought to explore how the playing area constrained external workloads (total distance covered, distance covered while walking, running, and sprinting, number of sprints, maximum sprint speed), internal load perceptions (rating of perceived exertion, RPE), and technical actions of passing (number of passes with dominant and non-dominant foot, and maximum passing speed) on players during 4v4 SSCGs aimed at maintaining and recovering ball possession. We expected to observe greater changes of external, internal loads and technical actions between playing area manipulations in U11 than in U15 and U23. Specifically, it is expected that U11 reveal higher external, internal loads and technical actions in largest playing areas in comparison to smallest due to their technical and tactical limitations to play. It is expected in U15 and U23 a similar tendency between playing areas as observed for U11 but with lower magnitude.

Methods

Participants

Fifty-two academy level participants from three different football team age groups participated in this study (U11, $n = 16$, age: 10.0 ± 0.7 y, body mass: 33.0 ± 2.34 kg, height: 141.0 ± 4.6 cm, years of experience: 1.0 ± 1.1 y; U15, $n = 18$, age: 14.0 ± 1.3 y, body mass: 58.0 ± 13.4 kg, height: 169.0 ± 10.1 cm, years of experience: 3.0 ± 1.2 ; U23, $n = 18$, age: 21 ± 1.60 y, body mass: 66.5 ± 10.1 kg, height: 174.5 ± 4.3 cm, years of experience: 6.5 ± 1.6 y). Two of U15 and U23 players suffered injuries during the subsequent weeks and were replaced by two new players of the team. All participants were part of the same team and experienced three weekly training sessions plus one game on weekends. The experimental protocol and investigation were approved by the local Institutional Research Ethics Committee and performed in accordance with the ethical standards of the Helsinki Declaration. For the U11 and U15 teams, written informed assent was obtained from the parents of the players. The same written informed consent procedure was also undertaken with the U23 participants.

Procedures

Participants performed in a series of 4-a-side SSCGs (Abrantes et al., 2012; Aguiar et al., 2013) on different playing area dimensions: Small playing area (20×15 m; playing density: 37.5 m²) (S), Medium playing area (25×20 m; playing density: 62.5 m²) (M) and Large playing area (30×25 m; playing density: 93.75 m²) (L) (Owen et al., 2004; Williams & Owen, 2007). Head coaches were present during data collection and assigned players into balanced teams, based on players' capabilities. All players were part of the same academy team and had three weekly training sessions plus one weekend game. The U23 team plays 11-a-side; the U15 team plays 8-a-side football, in half pitch; and the U11 team plays 5-a-side football on a futsal pitch. For that reason, players' positions were not considered

when forming teams, as some players performed matches in different positions throughout the season. The 4v4 SSCG was performed for four sets of four minutes each, with four minutes of active recovery between games (in a total of sixteen minutes of intermittent exercise for each SSCG; $4 \times 4 \text{ min} + 4 \text{ min}$ recovery time) (Halouani et al., 2014). Three training sessions were completed on three different days for each age category, with each session being played on the same playing area (day 1 on M; day 2 on S; and day 3 on L). Additionally, each session was performed in the same training session of the microcycle, with a week of difference. Before the beginning of each session, the players performed a general warm-up that included running at various intensities, joint mobilization and stretching, for ~20 mins duration. Due to the purpose of the exercise (maintaining and recovering ball possession), no goalkeeper or any type of goal or target was used. The coach did not intervene during the SSCG with any corrective feedback. If the ball went out of play, other strategically placed balls allowed an immediate restart from a pass.

Data collection

Data on the external workload variables were collected through a GPS system (Global Position System) included in the ZEPP Play Soccer system, which uses 2 Micro Electromechanical Systems (MEMS) sensors to measure motion and with a Bluetooth 4.0 Low Energy connectivity. Each player had a microchip (each with 2 internal sensors: 3-Axis Accelerometer + 3-Axis Gyroscope) attached to each of their gastrocnemius to record displacement data (Aroganam et al., 2019; Sasaki et al., 2019). Later, Zepp's computer software (version 1.6.0) was used to compute the values of total distance covered (m), distance differentiated by walking ($\leq 9 \text{ km/h}$), running ($9\text{--}18 \text{ km/h}$) and sprinting ($>18 \text{ km/h}$), number of sprints (n), maximum sprint speed (km/h), number of passing (ball contacts) with dominant and non-dominant foot (n), and maximum passing speed (km/h) (Owen et al., 2014). The internal load perceptions were measured using a Borg Scale CR10 to evaluate RPE. It was presented to participants after the end of each SSCG to ensure that the perceived effort was referred to that specific game (Coutts et al., 2009). RPE value was chosen because it correlates well with traditional ways of obtaining information on training intensity in SSCGs (Heart Rate and Blood Lactate) and it was used in this study to monitor exercise intensity and the effects of training stimuli (Coutts et al., 2009). For the technical actions of passing, it was counted when the ball travelled at least 5 metres using ZEPP Play Soccer system.

Statistical analysis

A descriptive analysis was performed using mean and standard deviations. The normality and homogeneity of the variances were tested, and the sample met the requirements for the assumption of normality. Afterwards, a one-way repeated measured analysis of variance (ANOVA) was conducted to evaluate differences in performance variables for each age group according to playing area dimension (i.e. small (S), medium (M) and large playing areas (L)). The statistical analysis was performed using the Statistical Package for Social Sciences software V24.0 (IBM SPSS Statistics for Windows, Armonk, NY: IBM Corp.), and the statistical significance level was set at $p < .05$.

Complementary, pairwise differences (S vs. M, S vs. L and M vs. L playing areas) for each age group were assessed via differences in group means expressed in raw units with 95% confidence limits (CL). All differences were assessed via standardized differences with pooled variance, derived from the mean and standard deviation of each variable, with 95% Confidence Intervals (95% CI). The magnitude thresholds for mean differences were 0.20, 0.60, 1.2, 2.0 and 4.0 for small, moderate, large, very large and extremely large, respectively (Cumming, 2012).

Results

Table 1 shows the descriptive analysis for all dependent variables across the different age groups performed in the S, M and L playing area conditions. Table 2 shows the effect of playing area conditions on the different performance variables for each age group. Table 3 indicates the pairwise differences, with $\pm 95\%$ confidence limits, between the playing area conditions for each age group. Finally, the magnitude of differences from the above-mentioned comparisons is presented in Figure 1.

Under-11 age group

Regarding external workload, significant differences were observed for all variables ($p < .001$) with exception of the max speed ($p > .05$) between playing area formats (see Table 2). In general, the L playing area promoted an increase in all external workload measures in comparison with S and M playing area formats with exception of distance cover while walking, which moderate decreased ($p < .01$ and $p < .001$) in this playing area format. Pairwise comparisons showed moderate to large differences between S vs. L and M vs. L playing areas (see Table 3 and Figure 1). Less differences were found when comparing S with M playing areas. Still, through pairwise comparisons, it was also possible to notice an increase, with a large effect size, on distance covered while sprinting ($p < .001$) and number of sprints sprinting ($p < .001$ and $p < .01$), and with a moderate effect size on distance cover while running ($p < .001$ and $p < .05$) and max speed, when comparing S vs L and M vs L. The internal load also revealed significant differences for RPE between playing area formats ($p < .01$). In opposition to results of external load, the L playing area was associated to a decrease in internal load measures in comparison with S playing area, with a small effect size, and M playing area, with a moderate effect size ($p < .01$). A small increase in internal load was also observed between S and M playing area formats. No significant differences were observed for technical actions. However, a small increase was observed for almost all variables when comparing S and L playing areas. In general, a small increase was observed for L playing area in all technical actions in comparison with S (max passing speed, passing number, and with dominant and non-dominant foot) and M (passing number and with dominant foot) playing areas formats. Additionally, M playing area also showed a small increased effect on passing with non-dominant foot, when comparing with S format. Therefore, larger playing areas promoted an increase of technical action repetitions compared with smaller formats ($L > M > S$).

Table 1. Descriptive analysis for all age groups, when playing in different playing areas.

Variables	U11			U15			U23		
	S	M	L	S	M	L	S	M	L
External Load									
Walking (m)	149.8 ± 23.2	156.2 ± 27.7	124.9 ± 27.1	194.5 ± 34.9	189.9 ± 37.1	183.4 ± 31	189.8 ± 28.4	189.7 ± 40.8	188.7 ± 25.9
Running (m)	116.6 ± 46.4	130.7 ± 58.4	175.6 ± 51.4	137.3 ± 56.9	131.8 ± 69	162.1 ± 66.4	174.6 ± 55.2	159.5 ± 71.9	174.5 ± 55.2
Sprinting (m)	5.9 ± 6	9.4 ± 7.5	30.9 ± 25.7	3.9 ± 6.7	10.3 ± 15.7	9.2 ± 10.4	3.4 ± 4.3	4.2 ± 5.6	5.7 ± 5.7
Sprint number (counts)	1 ± 0.9	1.4 ± 1.2	3.5 ± 2.4	0.4 ± 0.7	1.3 ± 1.5	1.3 ± 1	0.4 ± 0.6	0.4 ± 0.6	0.8 ± 0.7
Max speed (km/h)	13.9 ± 2.7	13.9 ± 1.8	15.9 ± 2	15 ± 2.2	17 ± 2.6	17.2 ± 2.9	16.8 ± 2.7	16 ± 2.7	17.1 ± 1.9
Internal Load									
RPE (a.u.)	3.9 ± 1.1	4.6 ± 1.8	3.5 ± 1	4.3 ± 1.1	4.7 ± 1.2	4.6 ± 1	4.7 ± 0.9	4 ± 1.1	4.9 ± 0.9
Technical actions									
Max passing speed (km/h)	32.9 ± 8.3	34.3 ± 8.1	35 ± 8.3	34.9 ± 6.6	35.5 ± 10.9	37.5 ± 7.5	37.8 ± 7.8	35.5 ± 7.5	37.1 ± 15.9
Passing number (counts)	8.7 ± 4.9	9.5 ± 4.6	11.3 ± 7.4	7.1 ± 3.1	8.2 ± 5	7.1 ± 2.7	8 ± 3.6	7.7 ± 4	6 ± 4.1
Dominant foot (counts)	6.9 ± 4	7 ± 3.4	8.8 ± 6.5	5.7 ± 2.4	6.4 ± 4.1	5.4 ± 2.1	6 ± 2.7	5.7 ± 2.4	4.7 ± 3.1
Non-dominant foot (counts)	1.8 ± 1.6	2.5 ± 1.9	2.5 ± 1.8	1.4 ± 1.5	1.7 ± 1.8	1.7 ± 1.4	2.2 ± 1.4	2.1 ± 2	1.3 ± 1.3

Table 2. Results of the one-way repeated measured analysis of variance for each age group considering the playing area effect.

Variables	U11			U15			U23		
	<i>F</i>	<i>p</i>	η^2p	<i>F</i>	<i>p</i>	η^2p	<i>F</i>	<i>p</i>	η^2p
External Load									
Walking (m)	11.53	<.001	.27	0.85	.43	.03	0.01	.99	.00
Running (m)	9.76	<.001	.24	2.00	.15	.06	0.62	.53	.02
Sprinting (m)	24.31	<.001	.44	2.84	.07	.08	1.89	.16	.06
Sprint number (counts)	22.58	<.001	.42	6.70	<.01	.18	5.41	<.01	.15
Max speed (km/h)	2.02	.15	.06	4.49	.02	.13	0.27	.76	.01
Internal Load									
RPE (a.u.)	5.97	<.01	.16	1.25	.29	.04	6.83	<.01	.18
Technical actions									
Max passing speed (km/h)	0.12	.89	.00	1.37	.26	.04	0.24	.73	.01
Passing number (counts)	2.02	.15	.06	0.87	.39	.03	2.79	.07	.08
Dominant foot (counts)	1.87	.17	.06	0.97	.37	.03	2.07	.14	.06
Non-dominant foot (counts)	1.54	.23	.05	0.28	.72	.01	2.91	.06	.09

Under-15 age group

Regarding external workload, the L playing area promoted an increase only in the number of sprints ($p < .01$) and max speed ($p < .05$) when compared to the S playing area, and the M playing area promoted an increase in max speed ($p < .05$) when compared to the S playing area ($S < M < L$) (see [Tables 2](#) and [3](#)). Pairwise comparisons showed a small increased effect on the distance covered while running and sprinting in larger playing areas ($S < M < L$), and a small decreased effect on the distance covered while walking when comparing S vs L playing areas (see [Table 3](#) and [Figure 1](#)). A moderate increased effect was observed for number of sprints ($p < .05$ and $p < .01$) and max speed ($p < .05$) when comparing S vs M and S vs L formats. No significant differences were found in terms of internal perceptions of workload. Although, pairwise comparisons presented small increased effects of RPE for larger playing areas (M and L) when comparing with S. No significant differences were also observed for technical actions of passing. However, and in contrast to the other age groups, pairwise comparisons showed small decreased effects in the M playing area for passing number and with dominant foot passing when comparing with L playing areas. Moreover, small increased effects were found in the L playing area for max passing speed when comparing with S and M playing areas, as also as for passing number and with dominant foot when comparing S with M playing areas. Overall, U15 age group demonstrated less differences between the analysed variables.

Under-23 age group

The analysis of external workload revealed statistical differences for the number of sprints ($p < .01$), with greater values on L playing areas when comparing with S and M playing areas ($S < M < L$) (see [Table 2](#)). It was also possible to observe, through pairwise comparison, a small increased effect on distance cover while running and sprinting, number of sprints ($p < .05$) and max speed when comparing M with L playing areas (see [Table 3](#) and [Figure 1](#)). An increase was equally noticed for distance covered while sprinting, with a small effect, and for number of sprints ($p < .05$), with a moderate effect, when comparing S with L playing areas. Furthermore, a decrease was found, with a small

Table 3. Post-hoc analysis and mean changes with $\pm 95\%$ confidence limits for each age group on players' performance measures.

Group comparison outcomes as: Mean changes with ±95% confidence limits										
Variables	U11			U15			U23			M vs. L
	S vs. M	S vs. L	M vs. L	S vs. M	S vs. L	M vs. L	S vs. M	S vs. L	M vs. L	
External Load										
Walking (m)	6.5; ±12.1	-24.8; ±8.7**	-31.3; ±13.6***	-4.7; ±14.9	-11.1; ±13.7	-6.4; ±14.9	-0.1; ±13.5	-1.1; ±12.4	-1; ±14.3	
Running (m)	14.1; ±25.4	59.1; ±19.0***	44.9; ±26.0*	-5.6; ±27	24.8; ±28.1	30.3; ±27.2	-15; ±24.1	0; ±26	15; ±29.1	
Sprinting (m)	3.6; ±2.5	25.0; ±7.4***	21.5; ±8.3***	6.5; ±5.2	5.3; ±4.0	-1.1; ±5.4	0.8; ±2	2.3; ±2.1	1.6; ±2.2	
Sprint number (counts)	0.4; ±0.4	2.5; ±0.7***	2.1; ±0.8**	0.9; ±0.5*	0.9; ±0.4**	0; ±0.5	0; ±0.2	0.4; ±0.2*	0.4; ±0.2*	
Max speed (km/h)	0.0; ±1.1	2.0; ±1.0	1.9; ±0.7	2.1; ±1.1	2.2; ±1.2*	0.1; ±1.1	-0.7; ±1.3	0.3; ±0.9	1.1; ±1	
Internal Load										
RPE (a.u.)	0.7; ±0.6	-0.3; ±0.4	-1.0; ±0.5**	0.4; ±0.4	0.3; ±0.5	-0.2; ±0.4	-0.6; ±0.4*	0.3; ±0.3	0.9; ±0.5*	
Technical Actions										
Max passing speed (km/h)	1.4; ±3.3	2.0; ±2.8	0.7; ±3.7	0.6; ±3.8	2.6; ±2.8	2; ±4	-2.2; ±3.4	-0.7; ±5.8	1.6; ±4.7	
Passing number (counts)	0.8; ±1.7	2.6; ±2.2	1.8; ±2.8	1.1; ±1.9	0; ±1	-1.1; ±1.7	-0.3; ±1.6	-2; ±1.5	-1.7; ±1.5	
Dominant foot (counts)	0.1; ±1.5	1.9; ±1.8	1.8; ±2.3	0.8; ±1.5	-0.3; ±0.9	-1; ±1.3	-0.3; ±1.1	-1.3; ±1.2	-1; ±1	
Non-dominant foot (counts)	0.7; ±0.6	0.7; ±0.8	0.0; ±0.9	0.3; ±0.7	0.2; ±0.5	-0.1; ±0.7	-0.1; ±0.7	-0.8; ±0.6	-0.7; ±0.7	

Abbreviations: U11 = under-11 age group; U15 = under-15 age group; U23 = under-23 age group; S = small playing area; M = medium playing area; L = large playing area; S vs M: mean differences and post-hoc analysis of M – S; S vs L: mean differences and post-hoc analysis of L – S; M vs L: mean differences and post-hoc analysis of L – M. * p < .05; ** p < .01; *** p < .001.

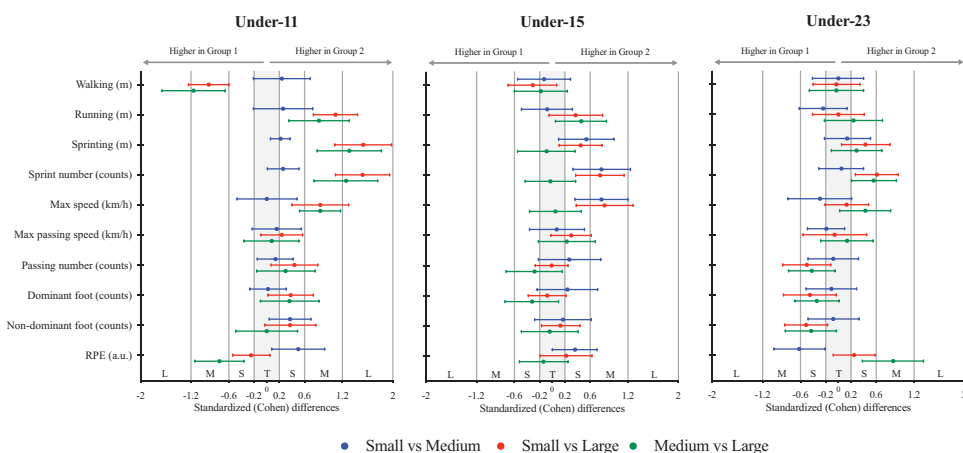


Figure 1. Standardized (Cohen) differences for considered variables according to the age group. Error bars indicate uncertainty in true mean changes with 95% confidence interval. Group 1 and Group 2 represent the comparison between playing area conditions (Small vs Medium; Small vs Big; Medium vs Big).

effect, on distance covered while running and max speed, when comparing S with M playing areas. The internal load also revealed statistical differences ($p < .01$) in the L playing area condition. A decrease, with a moderate effect, was also found for RPE scores in S playing area when comparing with M playing areas ($p < .05$); and an increase, with a small effect, in S vs L formats, and with moderate effect, in M vs L ($p < .05$). With regard to technical actions, no significant differences were observed for technical actions. Although, pairwise comparison showed significant effects for most of the variables when performing in smaller playing areas. Therefore, a decrease, with small effects, was found for passing number, and with dominant and non-dominant foot, when comparing S vs L and M vs L formats. Also, a small decrease effect was found on S vs M format for max passing speed.

Discussion

Several types of SSCG formats are commonly used by team sports coaches to constrain skill acquisition, conditioning and development of players and teams in training. The efficacy of this process depends on a deep understanding of the effects of manipulating practice tasks according to players' capabilities for action (Clemente et al., 2019). The aim of this study was then to evaluate the acute effects of playing area manipulations in ball possession SSCGs on physical effort and technical actions of players across three different age groups (U11, U15 and U23). The results are partially in line with our expectations. That is, the manipulation of playing area dimensions produces different effects on the external load and on the perception of exertion, only for the U11 age group. For these young players, large playing areas promoted higher external load, with lower internal loads in comparison to smaller playing areas. No differences were observed for technical actions. The U15 and U23 age groups revealed only subtle changes on the external load, with the number of sprints being higher for the large playing area in comparison to smaller ones. The U23 also revealed changes in internal load, with S and L playing areas promoting

higher RPE values than M playing area. As like U11, no differences were observed for technical actions.

In accordance with findings of Casamichana and Castellano (2010), as the playing area increases, actual playing time, total distance covered, and distance covered at different speeds (except walking) tend to increase. However, there is a need to understand how it occurs for different groups of players with different ages.

The U11 age group revealed great differences on external load between experimental conditions. It is well reported that younger players, while playing SSCGs, tend to pursuit the ball instead of keeping a balanced space coverage and to solve performance problems by individual actions rather than seeking for a collective solution (Almeida et al., 2013; Folgado et al., 2014; Lemes et al., 2019). Younger players still find it difficult to establish a proficient relationship with space, and according to changes in the space available to play, they cannot promote adaptative changes at a team level to maintain similar space time-equilibrium and consequently similar external load. Such difficulty to promote collective adaptations leads to the need of the players to adjust individually their actions according to the position of the ball and not based on the functionality of actions according to the specificities of game context (Travassos et al., 2018). In line with previous research, the manipulation of the playing area from S to L in ball possession tasks can be also used to promote a higher number of high-intensity actions and sprints for these young players (Hill-Haas et al., 2009).

As players become more experienced and competent, as the U15 and U23, they will be able to adjust their individual performance behaviours to the space covered and game dynamics of their own team, to the opposing team and the ball position, in order to explore the space available for play (Travassos et al., 2018). Thus, data may suggest that, as the players' age increases, the collective adaptations to maintain or recover ball possession in the available playing area may become more dominant and, by this it means, individual adaptations become less necessary. Contrarily to previous research (Hill-Haas et al., 2009) and to the results for U11 age group, no differences were observed between areas of play for U15 and U23 with exception to the sprint number. Such results come to reinforce that older players ensure collective balance in all the areas of play to maintain or recover ball possession, only increasing the number of sprints for L playing areas to ensure creation of spaces for play or interception of ball in the space (Vilar et al., 2014). Although more research is needed to support these suggestions and to understand the effects of different age categories on soccer players performance while practicing different formats of SSCGs. Specifically, it would be worthwhile to also analyse performance in accordance with different maturity levels of football players and not only considering age category.

With regard to the internal perception of workloads, most studies reported that larger playing areas have tended to be associated with more intense physical responses in participants (Casamichana & Castellano, 2010; Hill-Haas et al., 2009; Owen et al., 2004; Rampinini et al., 2007). However, others report that smaller playing areas make the exercises more intense (Tessitore et al., 2006). In our results, U23 reported higher RPEs on S, compared to M playing areas, completely opposite to U11, as this age group presented higher RPEs on M when compared to S; however, no differences were observed for U15 between areas of play.

These findings suggest that coaches should be aware that players in younger age groups seem not to perceive exercise in the same way as older players or that they cannot

work at the same performance levels of older players. This may occur due to an increase in tactical and physical capacity of players associated with development of action capabilities and exploration of opportunities of play, with older players reducing the running demands and younger players covering more distance while running when playing SSCGs in larger playing areas (Mendez-Villanueva et al., 2013). Specifically, data suggest that **young children should not be treated as** “mini-adults” as they have different needs and capacities, compared to older age players, in general. At this stage, it is important to use other tools that go beyond subjective internal load measures and external workload to monitor exercise intensity levels in younger players (e.g. heart rate, percentage of blood lactate or breathing rate). As a limitation to this investigation, further research would benefit from data using other indicators of intensity for different age groups while playing soccer SSCGs.

With regard to skill-based performance evaluation, no statistical differences were observed between playing areas for each age group. That is, the manipulation of playing areas in exercises of ball possession is not the most appropriate to promote different adaptations in technical skills, such as passing actions or using of dominant and non-dominant foot for passing. Despite this lack of statistical differences between playing areas for each age group, a higher frequency of passing actions was observed in the L playing area for the younger groups. The U11 age group performed more passing actions (including the dominant and non-dominant foot) on the L playing area. For the U23, the opposite tendency was observed, with more passing actions completed on the S playing area. These results indicate that larger playing areas greatly increased the affordances (opportunities) for younger players to perform a passing, given the reduced level of pressing to intercept the ball as the defenders had to cover more space and distance than on a S playing area. According to Turvey (1992), affordances are related to effectivities or capacities of individuals. The implication of this idea is that players at early stages of development should train on L dimension playing areas which have greater area per individual, since these performance spaces may allow more time for players to search for the most effective passing solution (Sgrò et al., 2018; Vilar et al., 2014). The emergent performance tendencies observed in the U23 age group also imply an affordance and effectivity relationship. The results are coincident with findings reported by Casamichana and Castellano (2010). They showed that the performance of soccer passing skills tended to increase in frequency, compared to other skills such as dribbling, as the playing area was reduced in dimensions. In fact, on S playing areas the distance between opponents gets smaller and more experienced performers may have increased the frequency of passing actions, relative to other actions, to adapt to these performance constraints. As players become more experienced, as more able they will be to discover new cooperation opportunities within the game. From a practical point of view, coaches should consider the possibility to reduce the available space in order to push the development of more experienced players in terms of technical execution and decision-making for passing. In young soccer players, SSCGs in small playing areas can then facilitate individual participation, while large playing areas will be more suitable for the development of tactical behaviours and decision-making (Clemente et al., 2019). Conversely, the U15 age group revealed small differences in passing actions across different game conditions. A possible explanation for these results is that players at this age group are usually in the middle of puberty and this may affect their ability to perform technical skills. Their bodies are

changing and it is usually a time when they want to try different skills and discover other ways to perform certain techniques, possibly by coping strategies (Malina, 2004). Participants in this age group may also be experiencing a change in growth and maturation with an increase in strength beginning to appear. These changes in physical properties could provide the impetus for the players to perform other skills such as dribbling with the ball past defenders, rather than passing more frequently at a younger age.

Small playing areas can then be used to work under pressure and develop decision-making processes, while large playing areas are suitable if the main goal is to work on tactical variability and physical stimulation, promoting technical development and being more physically demanded (Olthof et al., 2018). Further studies should continue to be conducted to understand the effects of practice task designs and age group variations in different types of SSCGs. For instance, investigators could compare differences in participant maturation levels, relative to variations in scaling of space and number of players involved, so that coaches can better understand the impacts of training designs on players development (Fitzpatrick et al., 2018; Woods et al., 2019).

Conclusion

To summarize, the findings from this study suggest how coaches can use space manipulation in SSCGs as an important task constraint to shape players' performance behaviour across different age groups. High-intensity running proved to be facilitated by the use of larger playing areas, especially for younger players. The findings suggest that coaches should carefully design and monitor the impact of high-intensity exercises, given that playing area manipulation presented a differential effect on the internal perception of workload across age groups. Space manipulation during SSCGs should also be considered as a major task constraint to facilitate and shape skill adaptations and development in learners. The use of smaller playing areas seems to favour the increase of passing actions in older players while, in turn, increases in playing area appear to alter the available time for younger players to practice skills without the major constraint of pressing. Interestingly, we also found a reduced impact of playing area manipulations on the different dependent variables for the U15 age group.

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